LAB 3 – ARCHITECTURE SELECTION & JUSTIFICATION

System: Self-Service Coffee Kiosk System

Architecture Selection

We chose **Layered Architecture** for the Self-Service Coffee Kiosk System.

Architectural Choice

The system is divided into separate components for user interaction, order processing, payment processing, data storage, and receipt printing. This provides a clear separation of responsibilities between different parts of the kiosk.

Reason 1 – Clear Separation of Responsibilities

The Touch Screen handles customer interaction, while the Order Manager handles order processing. The Payment Service manages credit-card payments, and the Menu & Pricing Database stores coffee and pricing information. Separating these responsibilities makes the system easier to understand, develop, and maintain.

Reason 2 – Easy Maintenance

Each component can be modified independently without changing the entire system. For example, the coffee menu or prices can be updated in the Menu & Pricing Database without changing the Touch Screen or Payment Service.

Security Advantage

The Payment Service can isolate credit-card processing from the other components. The Order Manager sends payment requests to the Payment Service without exposing sensitive payment information to the user-interface component. This provides better control over payment-related security.

Performance Benefit

The layered structure allows each component to perform a specific task efficiently. Menu and pricing information can be retrieved from the database, while payment processing and receipt printing are handled by their respective components. This reduces unnecessary processing in the user-interface component.

Components Used

1	Touch Screen / User Interface
2	Order Manager
3	Payment Service
4	Menu & Pricing Database
5	Receipt Printer

Conclusion: Layered Architecture is suitable because it provides clear separation of responsibilities, easier maintenance, controlled payment processing, and efficient communication between components.